

Importance Weighted Autoencoders on Dynamically Binarized MNIST

Reconstruction, DReG, and Progressive-Particle Training

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Deep Generative Models of Texts and Images

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Abstract

We reconstruct the one-stochastic-layer, $K = 50$ MNIST IWAE experiment of Burda et al. [2], then evaluate two meaningful extensions: DReG and progressive-particle DReG. The reconstruction reaches 87.804 test nats after 121 passes. DReG modestly improves likelihood to 87.480 nats but not KID. Progressive-particle DReG reaches 86.733 nats, KID 178.08 ± 21.76 , and trains in 14.0 minutes rather than about 71.5 minutes for fixed- K DReG. All numerical comparisons use seed 236; they are evidence for a practical profile, not a multi-seed superiority claim.

Part I

Paper Reconstruction

I.1 Original architecture

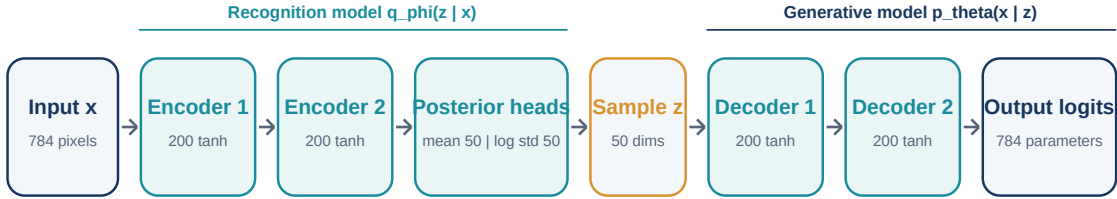
For a binarized image x , the model samples $z_{1:K} \sim q_\phi(z \mid x)$ and maximizes

$$\mathcal{L}_K(x) = \mathbb{E} \left[\log \frac{1}{K} \sum_{k=1}^K \frac{p_\theta(x, z_k)}{q_\phi(z_k \mid x)} \right]. \quad (1)$$

We minimize $-\mathcal{L}_{50}$ during training. Figure 1 and Table 1 give every architecture and training choice needed to reproduce the selected paper condition.

One-stochastic-layer IWAE: exact layer-by-layer architecture

The same architecture is used for the reconstruction, DReG, and progressive DReG.



Two 200-unit tanh layers in each network; a 50-dimensional diagonal Gaussian posterior; a factorized-Bernoulli decoder.

Figure 1: Source-backed architecture diagram. Both encoder and decoder use two 200-unit tanh hidden layers; the posterior is a 50-dimensional diagonal Gaussian and the decoder is Bernoulli.

I.2 Training set-up

The paper reports NLL in nats and active units. NLL is estimated here by $-\mathcal{L}_{5000}$ (lower is better); an active unit has posterior-mean variance above 0.01 across examples. The public MNIST dataset is downloaded by the project data module; the code and exact commands are in the repository README.

Table 1: Faithful reconstruction set-up and reported target.

Item	Paper / source code	Reconstruction
Data split	MNIST; source code reserves 400 examples	59,600 / 400 / 10,000
Training observations	dynamic Bernoulli binarization	matched
Architecture, latent size, particles	MLP, $z \in \mathbb{R}^{50}$, $K = 50$	matched
Initialization	Glorot; decoder log-odds output bias	matched
Optimizer	Adam, 10^{-3} , $\epsilon = 10^{-4}$	matched
Schedule	eight staged milestones through pass 3,280	matched milestones; run truncated at 121
Reported test NLL / active units	84.78 / 25	comparison target

I.3 Training versus validation

Training and validation use the shared $-\mathcal{L}_{50}$ objective; lower is better. Validation uses a deterministic reference binarization, while training uses dynamic binarization.

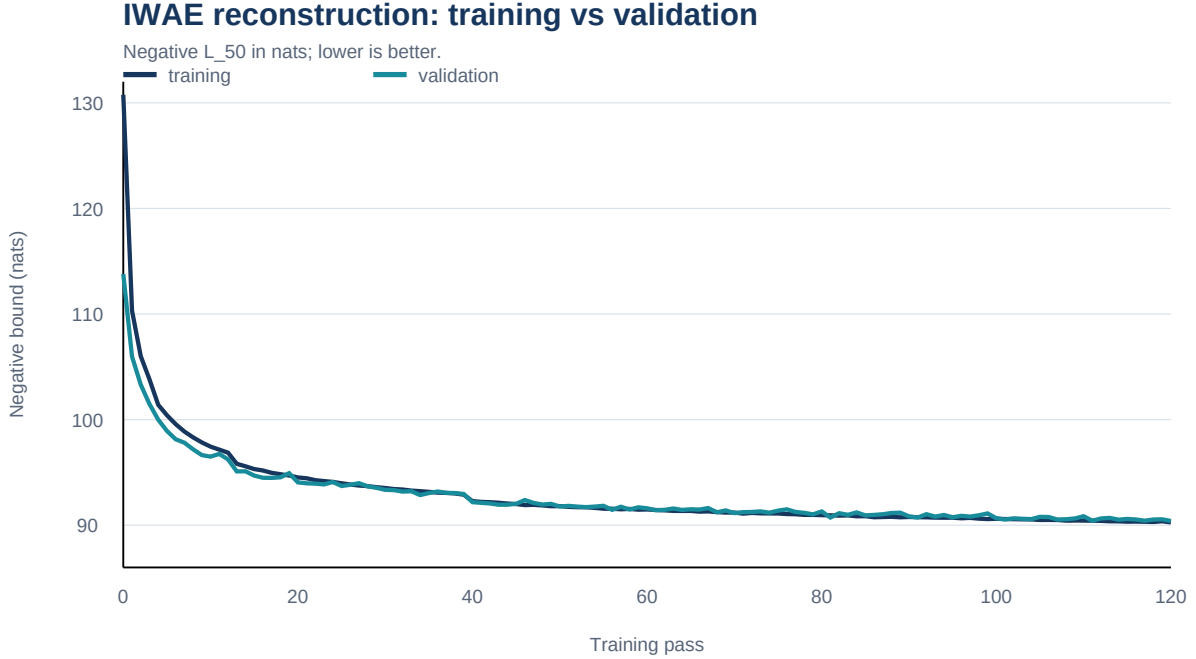


Figure 2: Reconstruction training versus validation $-\mathcal{L}_{50}$. The validation trace is selected on the same scale used for every condition.

I.4 Results against the paper

The reconstruction is 3.024 nats above the paper’s NLL after only 3.69% of the original pass budget, while reproducing the reported 25 active units.

Table 2: Paper result and reconstruction. NLL is the paper metric; lower is better.

Metric	Paper	Reconstruction (IWAE)
Training passes	3,280	121
Test $-\mathcal{L}_{5000}$ (nats)	84.78	87.804
Active units	25	25
KID	not reported	196.70 ± 23.05



Figure 3: Reconstruction: 64 identically seeded prior predictive decoder probabilities. Brightness is the Bernoulli mean.

Part II

DReG Improvement

II.5 Training set-up

DReG is the assignment’s meaningful loss/gradient modification: it retains the generative model and IWAE objective but replaces the encoder’s high- variance gradient estimator with the doubly reparameterized estimator of Tucker et al. [3]. The encoder surrogate is

$$-\sum_k \text{sg}(\tilde{w}_k)^2 \log w_k, \quad \tilde{w}_k = \frac{w_k}{\sum_j w_j}, \quad (2)$$

where sg stops gradients through the normalized weight. The decoder keeps the ordinary IWAE gradient. Data, architecture, seed, initialization, $K = 50$, batch size 20, Adam settings, 121-pass budget, and the validation/test procedures remain fixed, so the changed estimator is the intended experimental variable.

II.6 Training versus validation

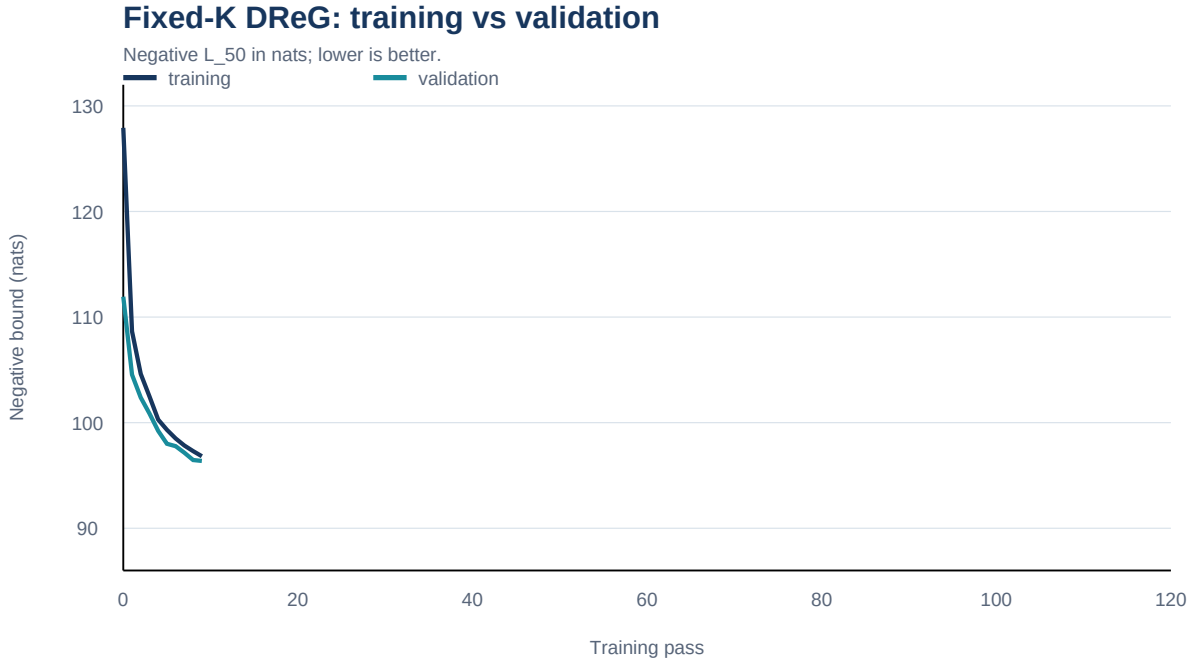


Figure 4: Fixed- K DReG training versus validation $-\mathcal{L}_{50}$. Both traces use the same axes as the reconstruction and progressive runs.

II.7 Results against reconstruction

Table 3: Matched seed-236 DReG comparison. The KID difference is not a run-to-run uncertainty estimate.

Metric	Reconstruction	Fixed- K DReG
Selected checkpoint pass	121	115
Best validation $-\mathcal{L}_{50}$	90.400	89.661
Test $-\mathcal{L}_{5000}$	87.804	87.480
Active units	25	26
KID	196.70 ± 23.05	206.80 ± 26.91
Training time (min)	74.2	approximately 71.5

DReG improves NLL by 0.323 nats but increases KID by 10.10 points. Thus it is a limited likelihood improvement, not a broadly superior model under this single-seed evidence.



Figure 5: Fixed- K DReG: 64 identically seeded prior predictive decoder probabilities. Brightness is the Bernoulli mean.

Part III

Progressive-Particle Improvement

III.8 Training set-up

This second improvement keeps DReG but changes the optimisation profile: batch size $20 \rightarrow 100$, particles $5 \rightarrow 10 \rightarrow 50$, and cosine learning-rate decay from 3×10^{-3} to 10^{-4} . Validation is always $K = 50$, making checkpoint selection directly comparable with fixed- K DReG. It is an optimisation extension, not a claim of paper-faithful training.

Progressive DReG training profile

The accelerated extension retains DReG and changes only the optimisation profile.

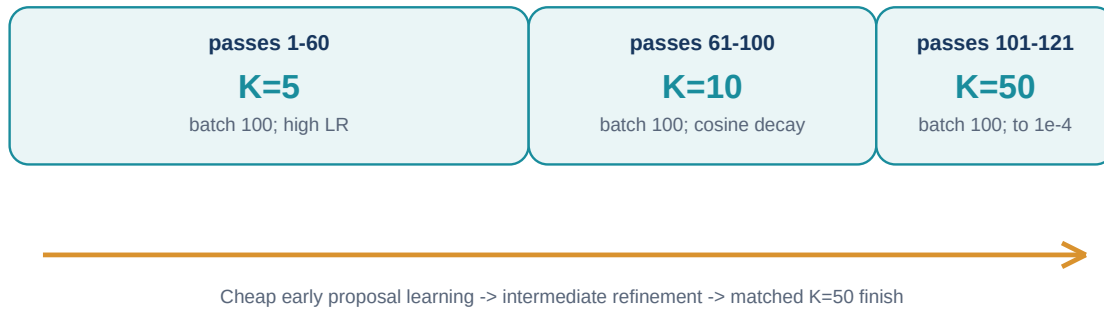


Figure 6: Exact accelerated training set-up. The final 21 passes use the same $K = 50$ objective as fixed- K DReG.

III.9 Training versus validation

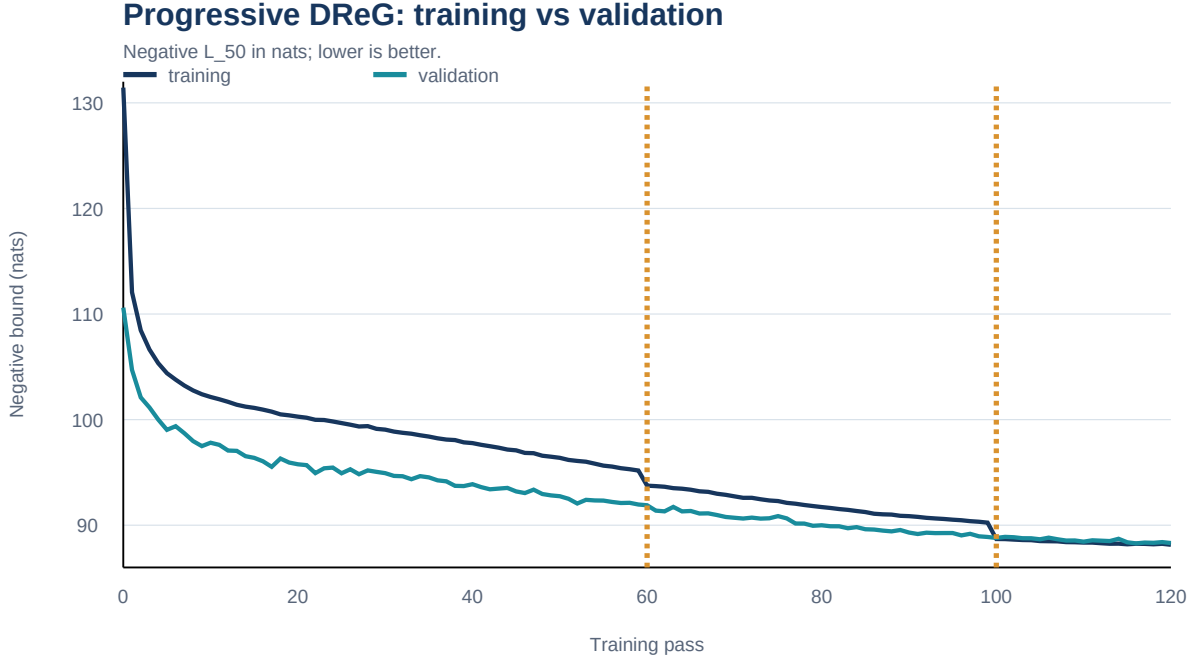


Figure 7: Progressive-particle DReG training versus validation $-\mathcal{L}_{50}$. The particle transitions occur after passes 60 and 100.

III.10 Results against reconstruction

KID is the required course metric in addition to the paper’s NLL. It is computed as Kernel Inception Distance [1] in a frozen 128-dimensional MNIST-classifier feature space. The shared classifier has 98.9% test accuracy; each result uses 10,000 real and generated images, 50 subsets of 1,000, and seed 236. Lower KID means closer feature distributions. The reported $\pm$ value is subset standard deviation, not replication uncertainty; the original paper did not report KID.

Matched seed-236 progressive-particle comparison against the reconstruction. Lower NLL,

Metric		Reconstruction	Progressive DReG
KID, and time are better.	Selected checkpoint pass	121	117
	Best validation $-\mathcal{L}_{50}$	90.400	88.278
	Test $-\mathcal{L}_{5000}$	87.804	86.733
	Active units	25	30
	KID	196.70 $\pm$ 23.05	178.08$\pm$21.76
	Training time (min)	74.2	14.0



Figure 8: Progressive-particle DReG: 64 identically seeded prior predictive decoder probabilities. Brightness is the Bernoulli mean.

Part IV

Final Comparison

IV.11 Table 4: completed results

Table 4 brings together the completed evidence. The paper value remains the long-run reference; the three 121-pass runs are matched seed-236 experiments. Lower NLL, KID, and time are better.

Table 4: Required final comparison of the paper, reconstruction, and both improvement iterations.

Metric	Paper	Reconstruction	Fixed- K DReG	Progressive DReG
Training passes	3,280	121	121	121
Test $-\mathcal{L}_{5000}$	84.78	87.804	87.480	86.733
Active units	25	25	26	30
KID	n/a	196.70 ± 23.05	206.80 ± 26.91	178.08 ± 21.76
End-to-end time (min)	n/a	74.2	approximately 71.5	14.0

IV.12 Validation comparison

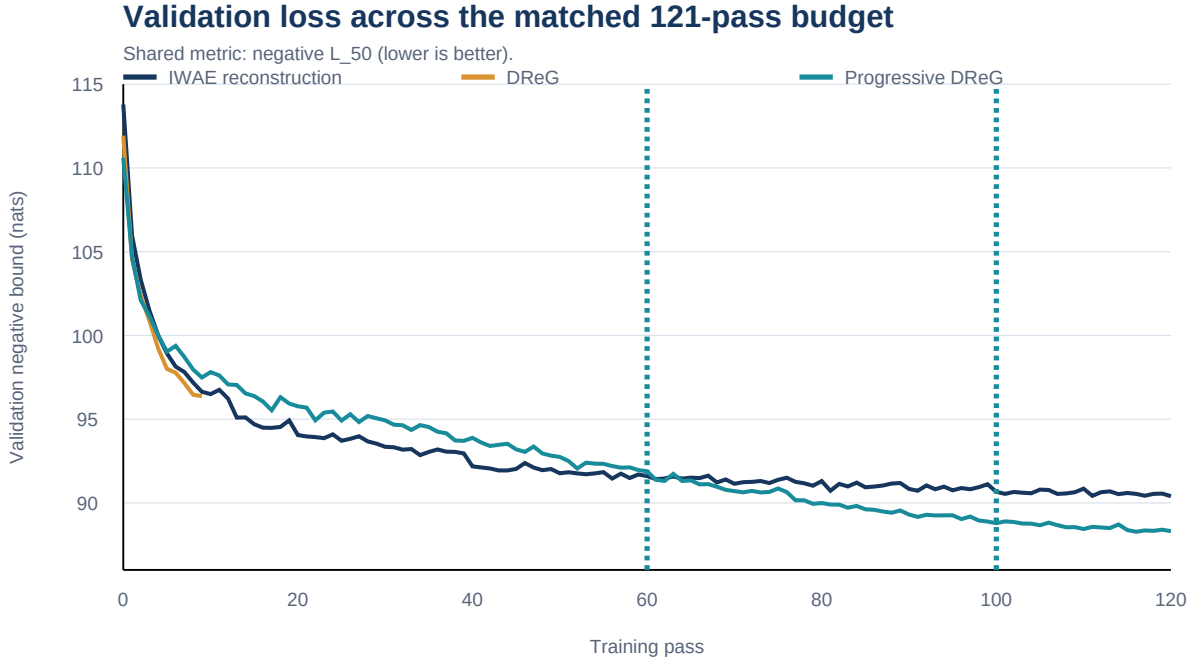


Figure 9: Final validation comparison of all completed 121-pass conditions. The vertical guides mark the progressive model’s particle transitions.

IV.13 Three generated sample grids

Prior samples from matched seed-236 checkpoints

Each panel uses 64 identically seeded latent draws; brightness is Bernoulli mean.



Figure 10: Three identically seeded prior-sample grids shown side by side: reconstruction, fixed- K DReG, and progressive-particle DReG. Brightness is the decoder’s Bernoulli mean.

IV.14 Discussion and lessons

- **What worked.** The original architecture and training core reproduce the paper’s 25 active units. DReG improves NLL, while the progressive profile improves NLL, KID, and elapsed time simultaneously.
- **What did not.** DReG alone worsened KID in this run. A truncated 121-pass reconstruction does not match the paper’s 3,280-pass NLL.
- **What we learned.** On this small IWAE, early $K = 50$ particles are expensive before the proposal is useful. Allocating particles over training is more consequential than changing the estimator alone.
- **Limitations.** All results use one seed. The fixed-DReG duration comes from interrupted-run artifacts, whereas progressive DReG is one uninterrupted run. Seeds 237 and 238, identical $\mathcal{L}_{5000}$ evaluation, KID, and a preregistered non-inferiority tolerance are needed before claiming reliable superiority.

Artifact availability and course-submission audit

The repository contains code for the reconstruction (`iwae train iwae`), the DReG and progressive improvements (`iwae train dreg` and `iwae train-fast`), $\mathcal{L}_{5000}$ evaluation, sample generation, the reusable MNIST feature classifier, and KID evaluation. The data module downloads public MNIST automatically from <http://yann.lecun.com/exdb/mnist/>. The relevant saved artifacts are:

- accelerated checkpoint: `outputs/fast-dreg/checkpoints/epoch-0116.ckpt`;
- shared KID classifier: `outputs/classifier/checkpoints/epoch-04.ckpt`.

These cover the reconstruction code, improved-model code, public dataset access, and PDF report required by the Part 2 brief.

References

- [1] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. In *International Conference on Learning Representations*, 2018. URL <https://arxiv.org/abs/1801.01401>.

- [2] Yuri Burda, Roger Grosse, and Ruslan Salakhutdinov. Importance weighted autoencoders. In *International Conference on Learning Representations*, 2016. URL <https://arxiv.org/abs/1509.00519>.
- [3] George Tucker, Dieterich Lawson, Shixiang Gu, and Chris J. Maddison. Doubly reparameterized gradient estimators for monte carlo objectives. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1810.04152>.